

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA5117

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO4: *Implement best security practices for IP, email, and web service using PGP, S/MIME for enhancing email Security.CO (BL3, BL6)*

Assessment Tool Weightage: 10%

QUESTIONS: 5

TOTAL MARKS: 50

Annexure A

Comparative Analysis

Code of Conduct:

*I Harshini R N (192421175) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Harshini R N

Annexure A
Comparative Analysis

1. Compare Authentication Header (AH) and Encapsulating Security Payload (ESP) in IPSec with respect to security services, performance overhead, and suitability for different network scenarios. Analyze which is more appropriate for a VPN deployment.
2. Compare transport mode and tunnel mode in IPSec architecture. Evaluate their effectiveness in securing host-to-host versus network-to-network communication.
3. Perform a comparative analysis of Pretty Good Privacy (PGP) and S/MIME in terms of key management, trust models, scalability, and enterprise usability. Recommend the most suitable approach for a corporate email system.
4. Compare rule-based and machine learning-based email spam detection techniques. Analyze their effectiveness in handling evolving spam patterns and minimizing false positives.
5. Compare manual key management and automated key management (e.g., using IKE) in IPSec. Evaluate their impact on security, scalability, and operational complexity in large organizations.

1. AH vs ESP in IPSec

Feature	AH	ESP
Authentication	Yes	Yes
Integrity	Yes	Yes
Confidentiality	No	Yes
Encryption	No	Yes
Replay protection	Yes	Yes
Performance overhead	Lower	Higher
Protection of IP payload	Yes	Yes
Protection of IP header	Stronger protection	Limited protection in transport mode
NAT compatibility	Poor	Better
Common usage	Integrity/authentication	VPN and confidential communication

Analysis

AH provides authentication and integrity but does not encrypt the data. Therefore, attackers can still read the packet contents.

ESP provides encryption and can also provide authentication and integrity. Although ESP has greater processing overhead, modern hardware and optimized cryptographic implementations reduce this impact considerably.

Suitability for VPN

ESP is more appropriate for VPN deployment because VPN traffic generally requires confidentiality as well as integrity and authentication. For site-to-site VPNs, ESP is normally used in **tunnel mode**.

Conclusion: AH is suitable when confidentiality is unnecessary and IP-packet authentication is important. ESP is the preferred choice for modern VPNs.

2. IPSec Transport Mode vs Tunnel Mode

Feature	Transport Mode	Tunnel Mode
Protects	IP payload	Entire original IP packet
IP header	Original header remains	Original header is encapsulated
Additional IP header	No	Yes
Overhead	Lower	Higher
Main application	Host-to-host	Network-to-network
VPN gateway support	Limited	Excellent
Hides internal IP addresses	No	Yes
Performance	Better	Slightly lower

Transport Mode

Original IP Header + Data
↓
IPSec Protection
↓
IP Header + Protected Data

Transport mode is suitable for **host-to-host communication**, where two individual systems communicate directly.

Tunnel Mode

Original IP Header + Data
↓
Encapsulated
↓
New IP Header + Protected Original Packet

Tunnel mode is suitable for **network-to-network communication**, particularly site-to-site VPNs.

Analysis

Transport mode has less overhead and can provide better performance, but it does not hide the original IP addresses.

Tunnel mode provides stronger network-level protection because the complete original packet is protected and a new outer IP header is added.

Conclusion: Use **transport mode for host-to-host communication** and **tunnel mode for network-to-network VPN communication**.

3. PGP vs S/MIME

Feature	PGP	S/MIME
Key management	User/decentralized	Centralized PKI
Trust model	Web of Trust	Certificate Authority
Scalability	Moderate	High
Enterprise management	More difficult	Easier
Certificate dependency	Usually no CA required	Uses X.509 certificates
Email client integration	Depends on implementation	Generally strong
Central policy control	Limited	Strong
Best suited for	Individual/technical users	Corporate environments

Key Management

PGP:

- Users generate and manage their own key pairs.
- Public keys can be exchanged directly.
- Trust is established through signatures and the Web of Trust.
- Large-scale key administration can become complicated.

S/MIME:

- Uses X.509 certificates.
- Certificates are issued by a CA or enterprise PKI.
- Certificate lifecycle can be centrally managed.
- Revocation and renewal can be automated.

Trust Model

PGP uses a **decentralized Web of Trust**, while S/MIME uses a **hierarchical CA-based trust model**.

Scalability and Enterprise Usability

S/MIME is generally easier to deploy across a large organization because administrators can centrally control certificates, policies, and trusted CAs.

PGP provides flexibility and decentralized control but can require more user involvement in key management.

Recommendation

For a **corporate email system**, **S/MIME is generally the better choice** because of:

- Centralized certificate management.
- Better enterprise integration.
- Easier key lifecycle management.
- Strong identity verification.
- Better scalability.
- Integration with corporate PKI.

Conclusion: Choose **S/MIME for large corporate environments** and PGP when decentralized trust and user-controlled key management are preferred.

4. Rule-Based vs Machine Learning-Based Spam Detection

Feature	Rule-Based	Machine Learning-Based
Detection method	Predefined rules	Learned patterns
Implementation	Simple	More complex
Adaptability	Low	High
New spam patterns	Poorer	Better
Maintenance	Manual rule updates	Model retraining
False positives	Can be high	Can be reduced with good training
Computational requirement	Low	Higher
Explainability	High	Usually lower
Scalability	Good	Very good with suitable infrastructure

Rule-Based Detection

Rule-based systems identify spam using predefined conditions such as:

- Suspicious keywords.
- Blacklisted senders.
- Suspicious URLs.
- Certain attachment types.
- Excessive capitalization.

Example:

```
IF sender is blacklisted  
OR subject contains "WIN FREE"  
OR message contains suspicious URL  
THEN classify as SPAM
```

Advantages:

- Simple.
- Fast.
- Easy to understand.
- Easy to audit.

Disadvantages:

- Attackers can modify spam patterns.
- Requires continuous manual updates.
- Can generate false positives.

Machine Learning Detection

ML systems learn patterns from previously classified emails.

Features can include:

- Email headers.
- Sender reputation.
- Message content.
- URL characteristics.
- Attachment information.
- Word frequency.
- Email structure.

Advantages:

- Adapts to changing spam patterns.
- Can identify previously unseen patterns.
- Can improve detection accuracy over time.

Disadvantages:

- Requires quality training data.
- More computationally expensive.
- Can be affected by adversarial or incorrectly labelled data.
- Less transparent than simple rules.

Conclusion

A **hybrid approach** is usually most effective:

```

Email
↓
Rule-Based Checks
↓
Header & Reputation Analysis
↓
Machine Learning Classifier
↓
Risk Score
↓
Spam / Legitimate / Quarantine

```

Rules provide fast, explainable detection while ML handles evolving spam patterns. Combining both can improve detection while reducing false positives.

5. Manual vs Automated IPSec Key Management

Feature	Manual Key Management	Automated Key Management (IKE)
Key generation	Manual	Automatic
Key distribution	Manual	Automatic
Scalability	Poor	Excellent
Human errors	High	Low
Key renewal	Manual	Automatic

Security	Lower for large deployments	Higher
Administration	Complex	Easier
Initial setup	Simple for small networks	More complex
Large enterprise suitability	Poor	Excellent
Fault recovery	Difficult	Better

Manual Key Management

In manual management, administrators configure encryption keys and IPSec parameters on each device.

For example:

```

Administrator
  ↓
Configure Key on Gateway A
  ↓
Configure Same Key on Gateway B
  ↓
Establish IPSec Communication

```

This can work for a small network but becomes difficult when hundreds or thousands of devices are involved.

Automated Key Management Using IKE

IKE automatically handles:

1. Peer authentication.
2. Security parameter negotiation.
3. Key exchange.
4. SA establishment.
5. Rekeying.
6. SA renewal.

```

Gateway A                                Gateway B
|                                         |
|----- IKE negotiation ---->|
|<----- Authentication ----|
|----- Key exchange ----->|
|                                         |
|===== IPSec SA Created =====|
|                                         |
|<===== Secure Traffic =====>|

```


Security

Manual keys can remain unchanged for long periods, increasing the risk if a key is compromised. Automated systems can regularly generate fresh session keys and renew SAs.

Scalability

Manual management becomes increasingly difficult as the number of VPN endpoints increases. IKE is much more suitable for large organizations because key negotiation and renewal are automated.

Operational Complexity

Manual management initially appears simple, but administrative effort grows rapidly with network size. IKE requires more configuration initially but substantially reduces ongoing administrative work.

Conclusion: For a large organization, **automated key management using IKE/IKEv2 is strongly preferred** because it provides better security, scalability, automatic rekeying, and lower operational complexity.